

on chemical and materials knowledge, such as those obtained from literature articles. In the future, it would be highly desirable for a chemically fine-tuned model to provide more insightful hypotheses and explanations, and even refer to specific references of prior published experiments to support them. These capabilities can potentially be within reach given the growing prevalence of powerful open-source LLMs and parameter-efficient fine-tuning.

In the current examples, LLMatDesign comes up with new materials designs from a limited set of modifications on the composition of a material. Nonetheless, this framework is general and can include more complex modifications which act not only on the composition space but also the structure space. Future work in this direction will focus on incorporating structural information when describing the material being modified, and also suggest modifications which directly act on the positions and lattice of the crystal structure. To this end, recent advances in multimodal LLMs can be applied here, where the atomic structure is considered to be an additional modality to be encoded in addition to the text modality.

3 Conclusion

In this work, we present LLMatDesign, a novel materials design framework powered by state-of-the-art LLMs that works directly with user-defined design requirements and constraints in natural language. It integrates computational tools for structure relaxation and property evaluation, incorporates internal chemical knowledge, and learns from previous iterations to function as an automated material design framework with high efficiency. Additionally, LLMatDesign quickly adapts to different tasks, target properties and design constraints by simply modifying the prompt. In our experiments, LLMatDesign consistently outperforms the baseline, demonstrating the effectiveness of the framework in developing new materials. Our work highlights the potential for fully automated AI-driven materials discovery that can be seamlessly integrated into autonomous laboratories in the future.

4 Methods

4.1 Large Language Models

Large language models (LLMs) are a class of machine learning models built on the transformer architecture [37]. By training on vast amounts of text data, these models can understand and generate text in a human-like manner. In this work, GPT-4o [34] refers to OpenAI’s `gpt-4o` model, which has a context length of 128K and a knowledge cutoff date of October 2023. Gemini-1.0-pro [35] refers to Google’s LLM with the same name, featuring a context length of 32K.

4.2 Machine Learning Force Field

Machine learning force fields (MLFFs) represent a significant advancement in computational chemistry and materials science. By utilizing state-of-the-art machine learning models and training on extensive datasets of atomic structures with energies, forces, and stresses, MLFFs can achieve high accuracy in predicting these properties, often rivaling *ab initio* methods such as density functional theory (DFT) [38]. More importantly, MLFFs provide these high-accuracy predictions with unprecedented computational efficiency, enabling the simulation of larger systems and longer timescales. In this study, we train a TorchMD-Net model [39] using the MatDeepLearn framework [40, 41]. The training dataset, curated from the Materials Project [6], comprises 187,687 crystal structures with associated energies, forces, and stresses. The model is trained for 400 epochs on a single Nvidia A100 80GB GPU.

4.3 Machine Learning Property Predictor

Similar to machine learning force fields (MLFFs), machine learning property predictors (MLPPs) leverage advanced machine learning models trained on large datasets to make fast and accurate predictions for specific

target properties. In this study, we train TorchMD-Net models to predict two separate properties: band gap and formation energy per atom. The datasets used are the `mp_gap` and `mp_form` datasets from the MatBench benchmark [42], containing 106,113 and 132,752 structures from the Materials Project [6], respectively. Each model is trained for 200 epochs on a single Nvidia A100 80GB GPU.

4.4 Modification of Material

Once LLMatDesign suggests a modification to achieve the user’s target objective, the material is modified accordingly. Specifically, as illustrated in Fig. 1, there are four types of modifications: **exchange**, **substitute**, **remove**, and **add**. Each modification is applied directly to an `ase.Atoms` object representing the material. For example, given the modification [`‘exchange’`, `‘Sr’`, `‘Ti’`], all Sr atoms in the material are replaced with Ti atoms and vice versa. After applying the modification, the structure undergoes relaxation using a machine learning force field (MLFF).

4.5 Modification of Material

The DFT calculations were performed using the Vienna Ab Initio Simulation Package (VASP)[43, 44]. All calculations followed the same settings specified by the "MPRelaxSet" in the Pymatgen library[45] used in Materials Project.

5 Data Availability

The authors declare that the data, materials and code supporting the results reported in this study are available upon the publication of this manuscript.

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SUPPLEMENTARY INFORMATION

A Prompt Templates for LLMatDesign

A slightly different prompt template to Fig. 2 is designed for Gemini-1.0-pro due to its inconsistency in generating standardized output.

LLMatDesign Prompt Template (Gemini-1.0-pro)

I have a material and its **<property>**. **<definition of property>**.

(**<chemical composition>**, **<property value>**)

You will be given a starting material to be modified. Try to achieve **<objective>**. Make an informed choice of modification based on the given material and past modifications and property values obtained after those modifications. Output a list for the suggested modification, and a string of the reason why you think it is a good modification to take to achieve **<objective>**. Make sure the modification is physically meaningful.

Material to be modified: **<chemical composition>**

Current property value: **<property value>**

<modification history>

Available modifications:

1. exchange: exchange two elements in the material
2. substitute: substitute one element in the material with another
3. remove: remove an element from the material
4. add: add an element to the material

Example output format:

1. ["exchange", "O", "N"], "some reason here"
2. ["substitute", "Ti", "Fe"], "some reason here"
3. ["add", "O"], "some reason here"
4. ["remove", "O"], "some reason here"

Figure A.1: Prompt template for LLMatDesign with Gemini-1.0-pro. Text placeholders in red angular brackets are specific to the task given to LLMatDesign. Text placeholders in blue angular brackets are optional and can be omitted if not needed.